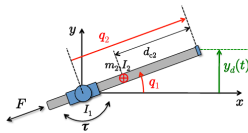


The RP planar robot in Fig. 1 with coordinates $q = (q_1, q_2)$ and parameters m_2, d_2, I_1 and I_2 defined therein, should execute a task defined by a time-varying trajectory $y_d(t) \in \mathbb{R}$ for the height of its end-effector.



Assuming as input command the joint velocity $\dot{q} \in \mathbb{R}^2$, determine the explicit expressions of the kinematic control laws that execute the task in nominal conditions, recover exponentially from any task error, and

- minimize $\frac{1}{2} \|\dot{q}\|^2$: which is the theoretical pitfall of this solution?
- minimize the weighted norm $\frac{1}{2} \dot{q}^T W \dot{q}$, with constant $W = \text{diag}\{w_1, w_2\} > 0$; what happens for very large ratios w_1/w_2 (in the limit $\rightarrow \infty$); and for $w_2/w_1 \rightarrow \infty$?
- minimize the kinetic energy $T = \frac{1}{2} \dot{q}^T M(q) \dot{q}$, being $M(q) > 0$ the robot inertia matrix.

$$\chi(q) = q_2 \sin(q_1) \Rightarrow J: (q_2 \cos(q_1) \quad \sin(q_1))$$

The First law is: $\dot{q} = \underset{\substack{\uparrow \\ \text{Feed Forward}}}{J^\#} \dot{\chi}_d + K \underset{\substack{\uparrow \\ \text{feedback}}}{J^\#} (\chi - \chi_d)$ with $K > 0$

The problem is that is minimizing $\dot{q}_1^2 + \dot{q}_2^2$ but these two quantities don't share the unit measure, q_1 is in rad, q_2 in meters.

to minimize $\|\dot{q}\|_W$ we consider

$$J_W^\# = W^{-1} J^T (J W^{-1} J^T)^{-1} \Rightarrow \dot{q} = J_W^\# (\dot{\chi}_d + K(\chi - \chi_d))$$

If $\frac{w_1}{w_2} \gg 1 \Rightarrow w_2 \gg w_1$, then the robot will follow χ_d by moving mainly q_2 , else if $w_1 \gg w_2$, it will move q_1 mainly (until $q_1 = \pi/2$) and then will move q_2 .

$$W^{-1} = \frac{1}{w_1 w_2} \begin{pmatrix} w_2 & 0 \\ 0 & w_1 \end{pmatrix} \quad J^T = \begin{pmatrix} q_2 c_1 \\ q_2 s_1 \end{pmatrix}$$

$$W^{-1} J^T = \frac{1}{w_1 w_2} \begin{pmatrix} q_2 c_1 w_2 \\ q_2 s_1 w_1 \end{pmatrix} \quad J W^{-1} J^T = \frac{1}{w_1} q_2^2 \Rightarrow (J W^{-1} J^T)^{-1} = \frac{w_1}{q_2^2}$$

$$\Rightarrow J_W^\# = \frac{1}{q_2 w_2} \begin{pmatrix} c_1 w_2 \\ s_1 w_1 \end{pmatrix} \Rightarrow \dot{q} = \frac{\dot{\chi}_d + K(\chi - \chi_d)}{q_2 w_2} \begin{pmatrix} c_1 w_2 \\ s_1 w_1 \end{pmatrix}$$

$$T = \frac{1}{2} I_1 \dot{q}_1^2 + \frac{1}{2} I_2 \dot{q}_1^2 + \frac{1}{2} m_2 (\dot{q}_2^2 + (q_2 - d_2)^2 \dot{q}_1^2) \quad \dot{q}_2^2 + q_1$$

$$M_{11} = I_1 + I_2 + m_2 (q_2 - d_2)^2 \quad M_{22} = m_2 \quad \text{So}$$

$$\Rightarrow \dot{q} = \frac{\dot{\chi}_d + K(\chi - \chi_d)}{q_2 m_2} \begin{pmatrix} c_1 m_2 \\ s_1 (I_1 + I_2 + m_2 (q_2 - d_2)^2) \end{pmatrix}$$

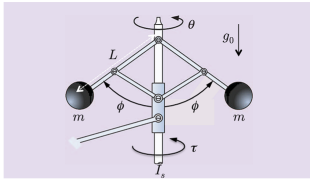


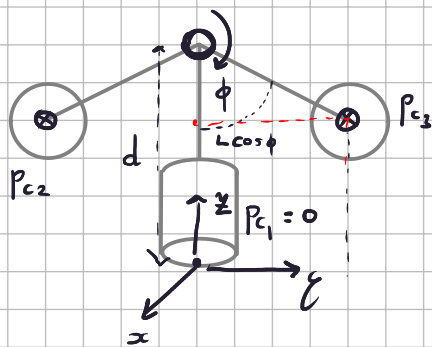
Figure 2: The Boulton-Watt governor and a scheme with definition of parameters and variables.

Figure 2 shows a picture and a simplified scheme of the famous Boulton-Watt centrifugal governor, a system invented to regulate the rotational speed of a steam engine by a mechanical leverage (feedback) opening a valve that provides steam under pressure to the engine. We consider here only the so-called *open-loop* dynamic behavior of the system, under the action of an external torque $\tau \in \mathbb{R}$ applied to the main rotating shaft.

Assume that:

- the main shaft has an inertia I_s around its rotation axis
- the two balls have identical mass m that is concentrated at the end of a link of length L
- the links and all other linkages have negligible masses
- a viscous friction torque with coefficient $f_v > 0$ is acting on the main shaft
- all other frictional effects are negligible.

Derive the complete dynamic model of this system using a Lagrangian formalism. Assuming knowledge of the geometric parameter L , provide a linear parametrization of the dynamics in terms of its dynamic coefficients. Find the value of the constant torque τ_1 to be applied for sustaining a steady-state rotation at a given angular speed $\Omega > 0$. Finally, design a nonlinear feedback for τ so as to achieve partial feedback linearization of the system, i.e., exact linearization by feedback of only part of the closed-loop dynamics, in this case of one of the two coordinates.

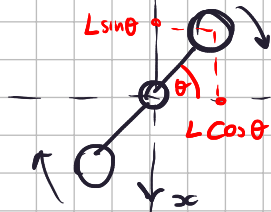


$$p_{c3} = L (\sin \theta, \cos \theta, \cos \phi)$$

$$p_{c2} = L (\sin \theta', \cos \theta', \cos \phi)$$

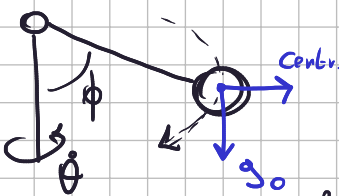
$$\theta' = \theta + \pi$$

$$\Rightarrow p_{c2} = L (-\sin \theta, -\cos \theta, \cos \phi)$$



Since the balls are considered point mass, we consider $T = \frac{1}{2} m v^2$

What forces acts on ϕ ? The centrifugal and gravitational.



Practically this is an under actuated robot.



$$q_1 = \theta \quad q_2 = \phi$$

$$\begin{pmatrix} \alpha & \beta & \gamma & \theta \\ \pi/2 & 0 & d_1 & q_1 \\ 0 & L & 0 & q_2 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & -L & 0 & -q_2 \end{pmatrix}$$

$${}^0A_1 = \begin{pmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1A_2 = \begin{pmatrix} c_2 & -s_2 & 0 & Lc_2 \\ s_2 & c_2 & 0 & Ls_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1A_3 = \begin{pmatrix} c_2 & s_2 & 0 & -Lc_2 \\ -s_2 & c_2 & 0 & Ls_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0A_2 = \begin{pmatrix} c_1 c_2 & -c_1 s_2 & s_1 & Lc_1 c_2 \\ s_1 c_2 & -s_1 s_2 & -c_1 & Ls_1 c_2 \\ s_2 & c_2 & 0 & Ls_2 + d_1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad {}^0A_3 = \begin{pmatrix} -Lc_1 c_2 \\ -Ls_1 c_2 \\ Ls_2 + d_1 \\ 0 \end{pmatrix}$$

$$\Rightarrow v_{c2} = \begin{cases} -Ls_1 c_2 \dot{q}_1 - Lc_1 s_2 \dot{q}_2 \\ Lc_1 c_2 \dot{q}_1 - Ls_1 s_2 \dot{q}_2 \\ Lc_2 \dot{q}_2 \end{cases}$$

$$v_{c3} = \begin{cases} Ls_1 c_2 \dot{q}_1 + Lc_1 s_2 \dot{q}_2 \\ -Lc_1 c_2 \dot{q}_1 + Ls_1 s_2 \dot{q}_2 \\ Lc_2 \dot{q}_2 \end{cases}$$

$$w_1 = \dot{q}_1$$

$$v_{c1} = 0$$

$$\frac{2}{m} T_2 = L^2 c_2^2 \dot{q}_1^2 + L^2 \dot{q}_2^2 \quad \text{is identical to } T_3$$

$$T = \frac{1}{2} I_s \dot{q}_1^2 + \frac{1}{2} m [2(L^2 c_2^2 \dot{q}_1^2 + L^2 \dot{q}_2^2)]$$

$$M_{11} = I_s + 2mL^2 c_2^2$$

$$M_{22} = 2mL^2$$

$$= e_1 + e_2 c_2^2 L^2$$

$$= e_2 L^2$$

$$M = \begin{pmatrix} e_1 + e_2 c_2^2 & 0 \\ 0 & e_2 \end{pmatrix}$$

$$e_2 = 2m$$

$$U = 2mg_0(LS_2 + d_1) \Rightarrow$$

$$g = \begin{pmatrix} 0 & g_0[2mLc_2 + 2md_1] \end{pmatrix} = \begin{pmatrix} 0 & g_0e_3 + e_4^L c_2 \end{pmatrix} = \begin{pmatrix} 0 & g_0e_2Lc_2 + e_3 \end{pmatrix}$$

$$C_1 = \frac{1}{2} \left(\frac{\partial M_1}{\partial \dot{q}_1} + \left(\frac{\partial M_1}{\partial \dot{q}} \right)^T - \frac{\partial M}{\partial \dot{q}_1} \right) = -e_2 S_2 C_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow C_1 = \dot{q}^T C \dot{q} = -2e_2 S_2 C_2 \dot{q}_1 \dot{q}_2$$

$$C_2 = -\frac{1}{2} \frac{\partial M}{\partial \dot{q}_2} = \begin{pmatrix} e_2 S_2 C_2 & 0 \\ 0 & 0 \end{pmatrix} = e_2 S_2 C_2 \dot{q}_1^2 \quad \text{the model is} \quad e_5 = S_1$$

$$\begin{cases} (e_1 + e_2 C_2^2) \ddot{q}_1 - 2e_2 S_2 C_2 \dot{q}_1 \dot{q}_2 + e_4 \dot{q}_1 = \tau \\ e_2 \ddot{q}_2 + e_2 S_2 C_2 \dot{q}_1^2 + e_3 + e_4 C_2 = 0 \end{cases} \quad \downarrow \text{we know } L$$

$$\begin{cases} (e_1 + e_2 L^2 C_2^2) \ddot{q}_1 - 2L^2 e_2 S_2 C_2 \dot{q}_1 \dot{q}_2 + e_4 \dot{q}_1 = \tau \\ e_2 L^2 \ddot{q}_2 + e_2 L^2 S_2 C_2 \dot{q}_1^2 + g_0 e_3 + g_0 e_2 L C_2 = 0 \end{cases} \Rightarrow$$

$$\begin{pmatrix} \ddot{q}_1 & L^2 C_2^2 \ddot{q}_1 - 2L^2 S_2 C_2 \dot{q}_1 \dot{q}_2 & 0 & \dot{q}_1 \\ 0 & L^2 \ddot{q}_2 + L^2 S_2 C_2 \dot{q}_1^2 + g_0 L C_2 & g_0 & 0 \end{pmatrix} e = \begin{pmatrix} \tau \\ 0 \end{pmatrix}$$

$$\ddot{q}_1 (e_1 + e_2 L^2 C_2^2) = 2L^2 e_2 S_2 C_2 \dot{q}_1 \dot{q}_2 - e_4 \dot{q}_1 + \tau \Rightarrow \text{FBL}$$

$$\tau = -2L^2 e_2 S_2 C_2 \dot{q}_1 \dot{q}_2 + e_4 \dot{q}_1 + (e_1 + e_2 L^2 C_2^2) a \Rightarrow \ddot{q}_1 = a$$

Consider the design of impedance control laws and force control laws for the 1-dof example, shown in Fig. 3, namely a single mass m that moves on a frictionless horizontal plane under the action of a commanded force $f \in \mathbb{R}$ and of a contact force $f_c \in \mathbb{R}$.

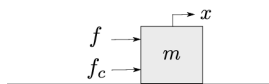


Figure 3: A mass m subject to a commanded force f and a contact force f_c .

In particular:

- The impedance controllers should work with a generic time-varying, smooth position reference $x_d(t)$, either with or without the use of a load cell that can measure the contact force f_c . Illustrate the properties of the obtained closed-loop systems.
- What happens when $x_d(t)$ degenerates to a constant? What happens during free motion, when $f_c = 0$?
- For $m = 5$ [kg], design the control parameters of the impedance law so that the dynamics of the position error $e = x_d - x$ in the closed-loop system is characterized by a pair of asymptotically stable complex poles with natural frequency $\omega_n = 10$ [rad/s] and critical damping ratio $\zeta = 0.7071$.
- On the other hand, the force controllers should be able to regulate the (measured) contact force f_c to a constant value f_d , using any combination of desired force feedforward and force error feedback. Illustrate the properties of the obtained closed-loop systems.
- What happens during free motion, when $f_c = 0$ and a constant contact force f_d is desired?

The dynamic model is $m\ddot{x} = f + f_c$, we want to achieve:

$$m_r(\ddot{x} - \ddot{x}_d) + b_r(\dot{x} - \dot{x}_d) + k_r(x - x_d) = f_c \quad \text{by imposing a}$$

suitable control force f . First we apply FBL:

$$f = m\ddot{x} - f_c \Rightarrow \ddot{x} = \ddot{x}_d, \quad \text{we want}$$

$$m_r\ddot{x} = f_c + m_r\ddot{x}_d - b_r(\dot{x} - \dot{x}_d) - k_r(x - x_d) \Rightarrow$$

$$\text{we choose } \ddot{x} = \frac{1}{m_r}(f_c + m_r\ddot{x}_d - b_r(\dot{x} - \dot{x}_d) - k_r(x - x_d)) \Rightarrow$$

$$f = \frac{m}{m_r}(f_c + m_r\ddot{x}_d - b_r(\dot{x} - \dot{x}_d) - k_r(x - x_d)) - f_c$$

Notice how if $m = m_r$, f_c cancel and the load cell is not required. If $f_c = 0$ we have:

$$m_r\ddot{x} = m_r\ddot{x}_d - b_r(\dot{x} - \dot{x}_d) - k_r(x - x_d) \quad \text{PD controller + feedforward}$$

$$\text{if } \ddot{x}_d = 0 \Rightarrow$$

$$m_r\ddot{x} = f_c - b_r\dot{x} - k_r(x - x_d) \quad \text{the system will reach an equilibrium } x_e:$$

$$m_r s^2 x = f_c - b_r s x - k_r x + k_r x_d \Rightarrow$$

$$x(m_r s^2 + b_r s + k_r) = f_c + k_r x_d$$

$$x = \frac{f_c + k_r x_d}{m_r s^2 + b_r s + k_r} \quad \lim_{t \rightarrow \infty} x = \lim_{s \rightarrow 0} x = \frac{f_c + k_r x_d}{k_r} = x_e \quad \text{steady state.}$$

$$\text{and } \lim_{k_r \rightarrow \infty} x_e = x_d.$$

$$\text{Reconsider } e = x_d - x \quad \dot{e} = -\dot{x} \quad \ddot{e} = -\ddot{x}$$

$$m_r\ddot{x} = f_c - b_r\dot{x} - k_r(x - x_d) \Rightarrow$$

$$-m_r\ddot{e} = f_c + b_r\dot{e} + k_r e \Rightarrow$$

$$m_r\ddot{e} + b_r\dot{e} + k_r e + f_c = 0 \quad s^2 + \frac{b_r}{m_r}s + \frac{k_r}{m_r}$$

$$e(s^2 m_r + b_r s + k_r) = -f_c \quad \begin{cases} \omega_n = \sqrt{\frac{\frac{1}{2}k_r}{2}} = \sqrt{k_r/10} = 10 \Rightarrow k_r = 1000 \\ \zeta = \frac{\frac{1}{2}b_r}{\sqrt{\frac{1}{2}k_r}} = \sqrt{2}/2 \Rightarrow b_r = 50\sqrt{2} \end{cases}$$

$$e = -\frac{f_c}{s^2 m_r + b_r s + k_r} = -\frac{\frac{1}{2}f_c}{s^2 + \frac{1}{5}b_r s + \frac{1}{5}k_r}$$

